Kernix 0.11.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

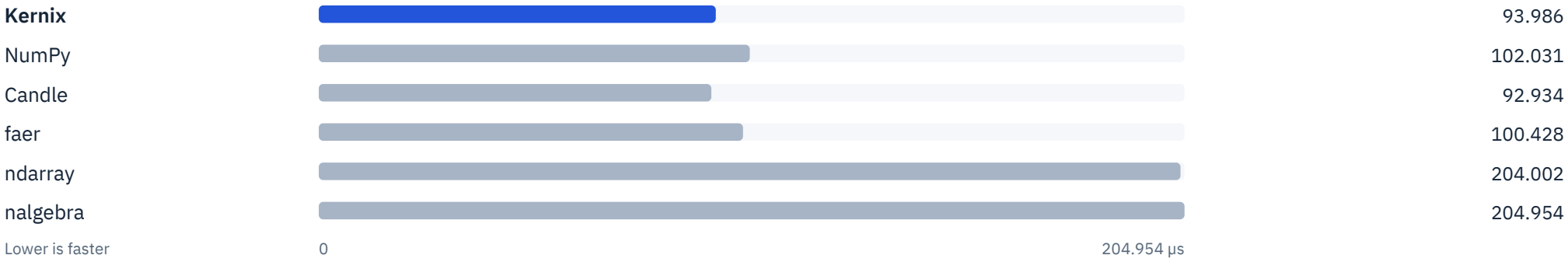
34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



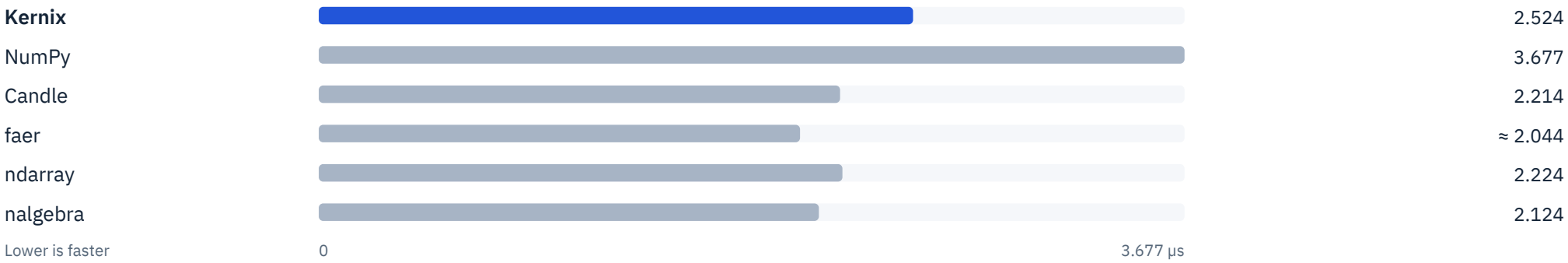
Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

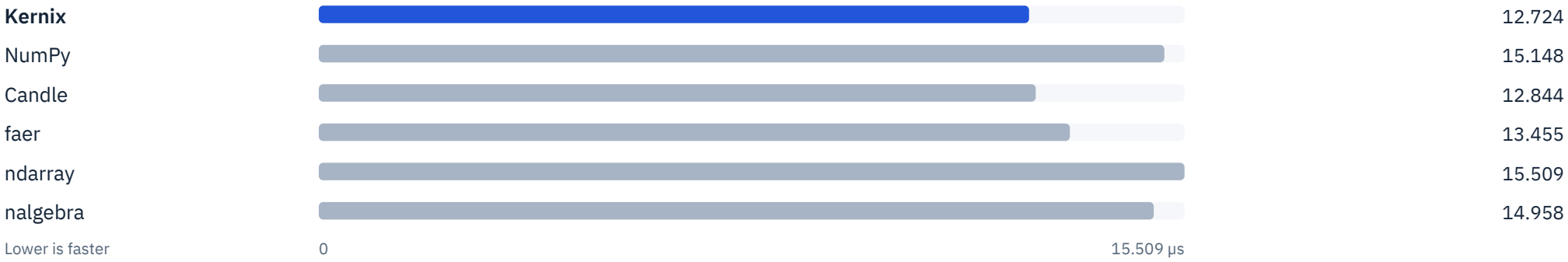
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



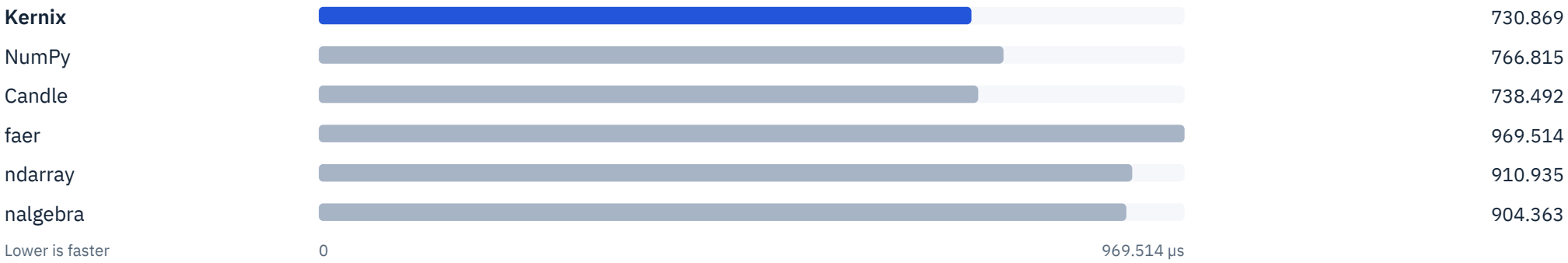
Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

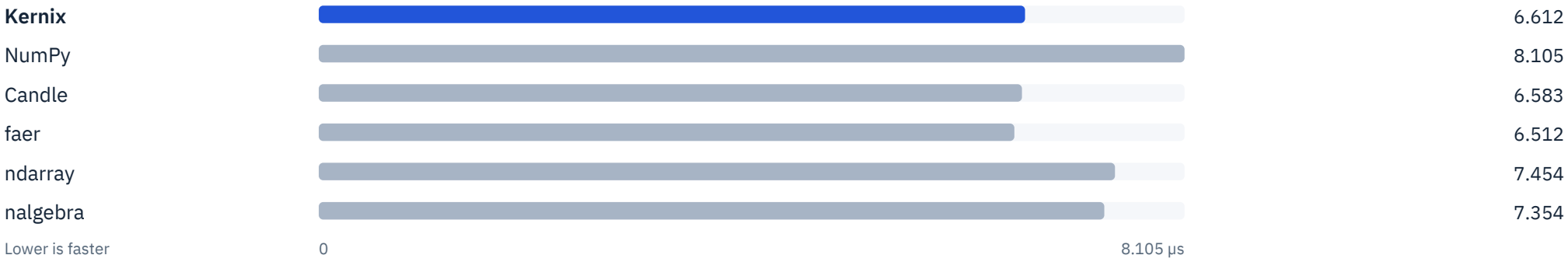
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



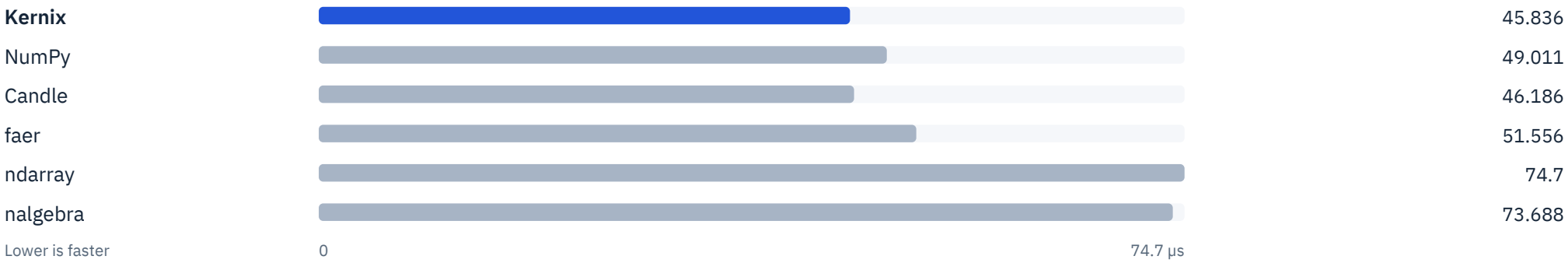
Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

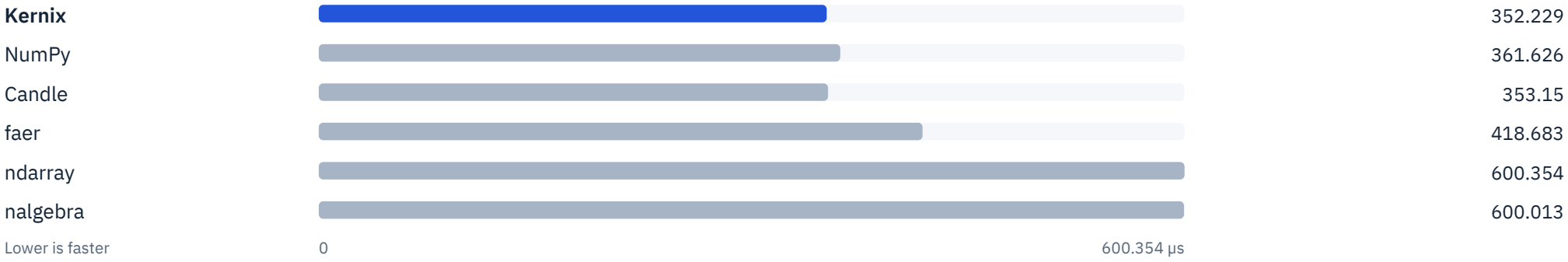
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

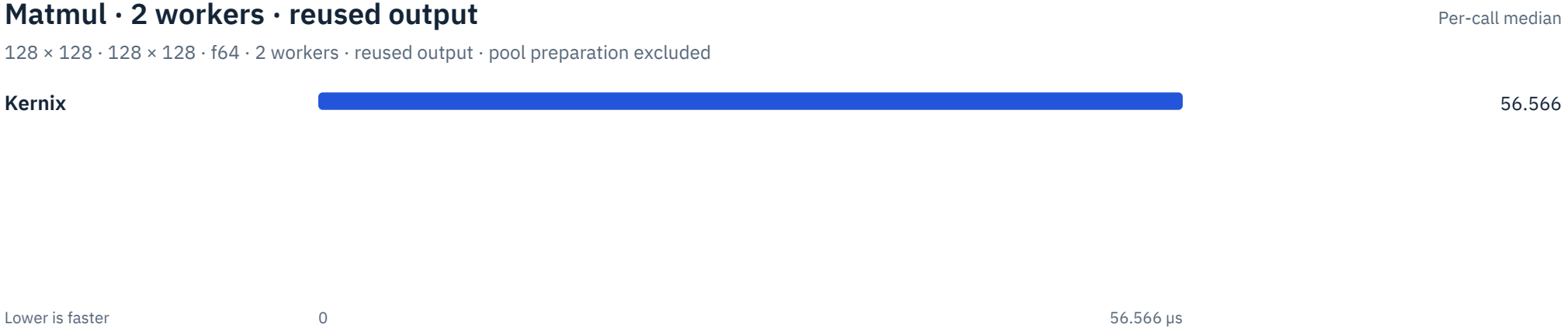
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

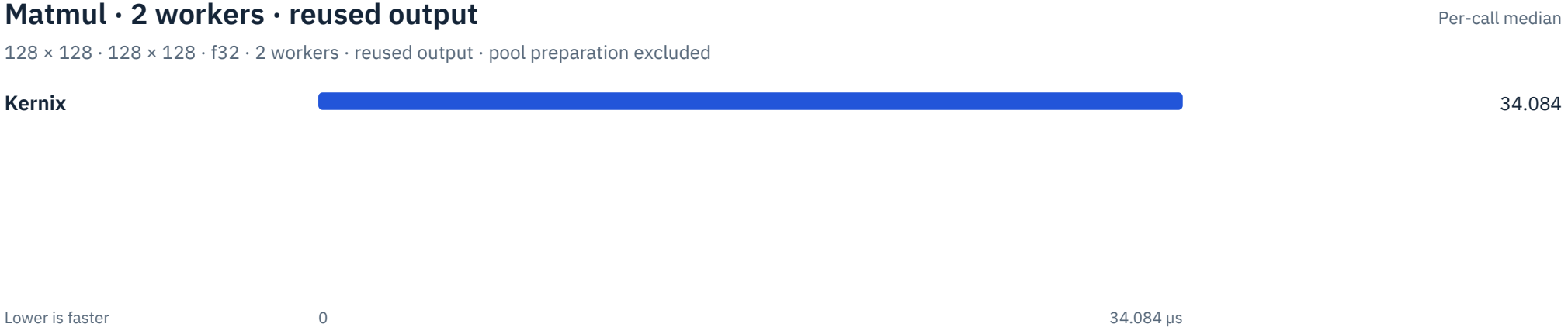
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



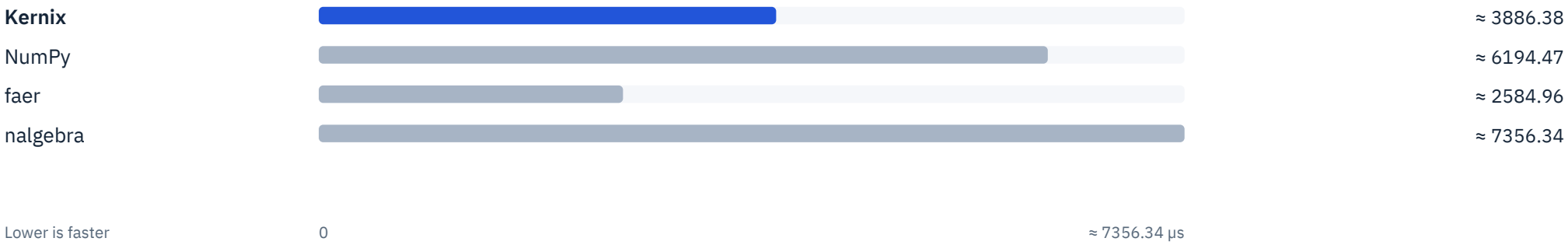
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

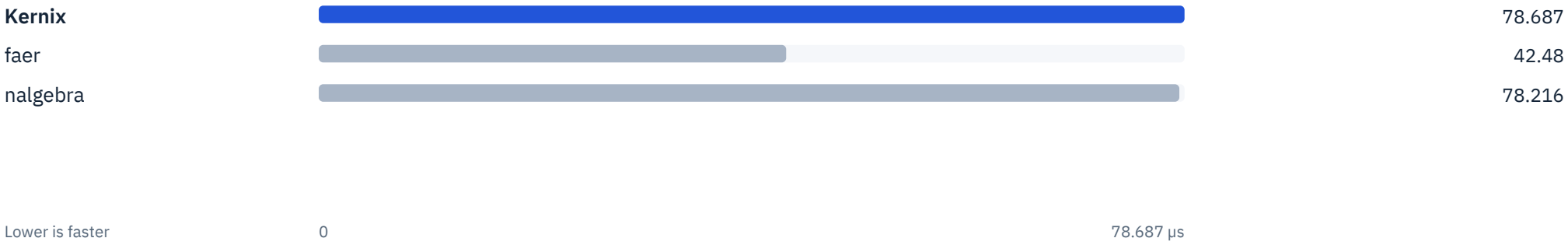
Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



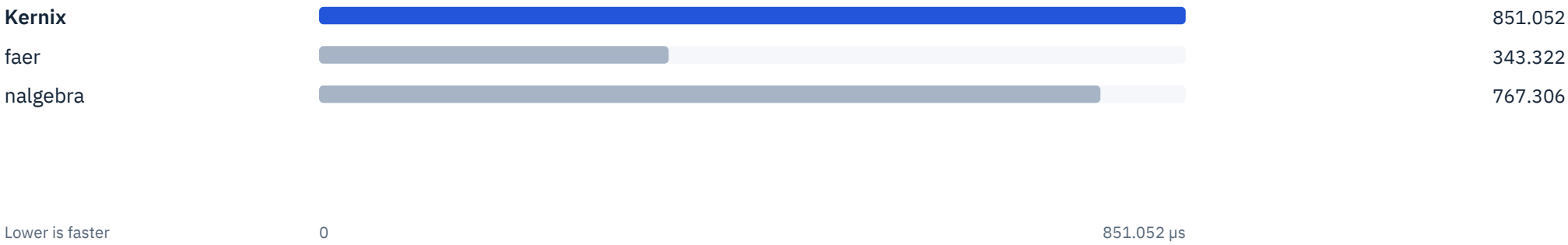
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

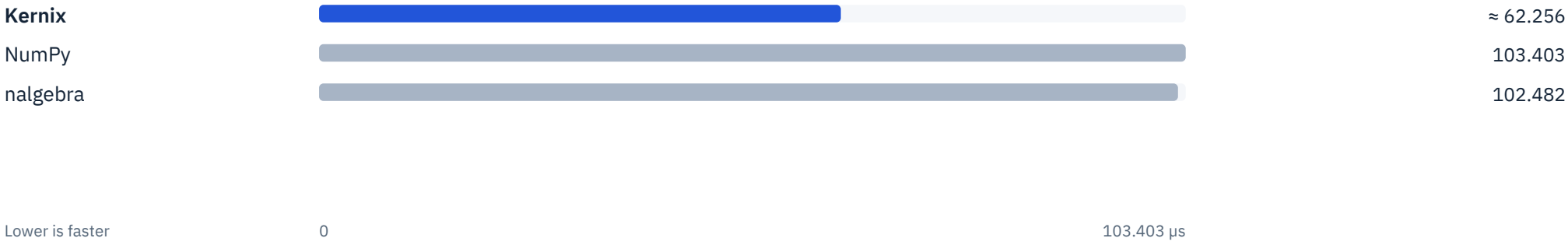
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



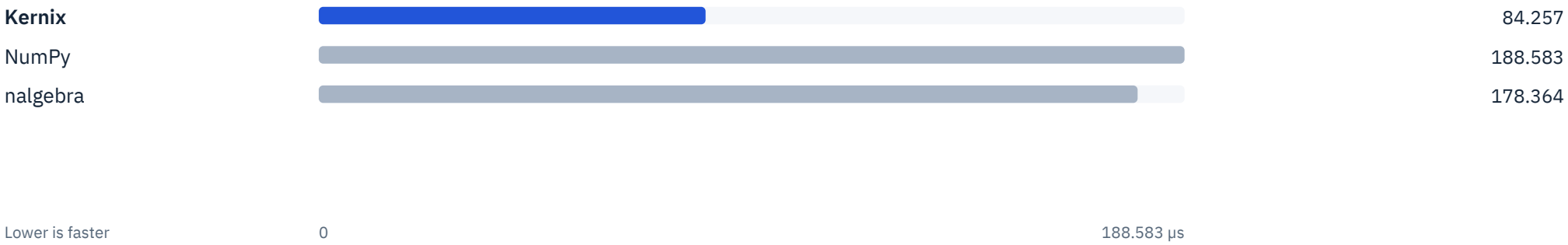
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

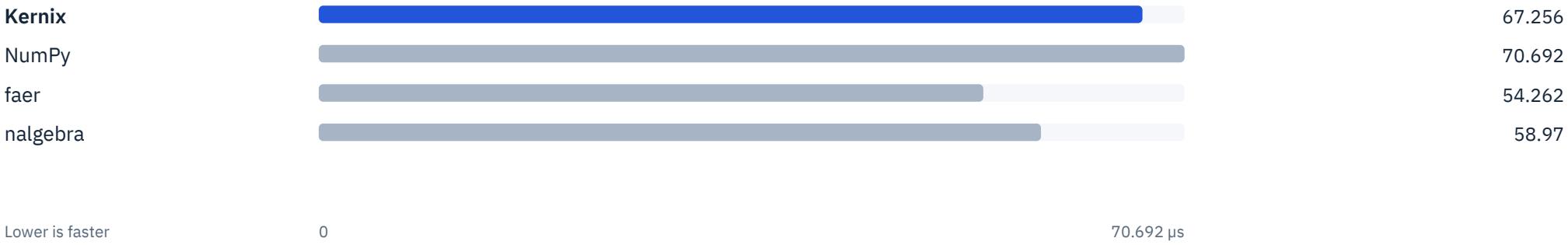
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



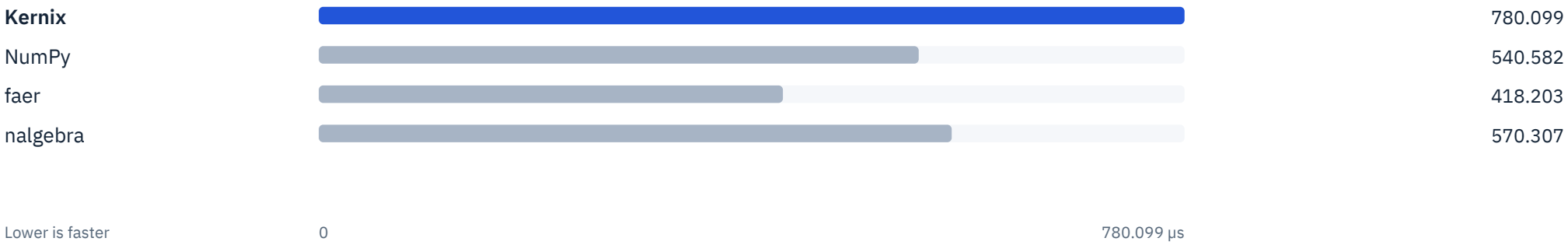
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

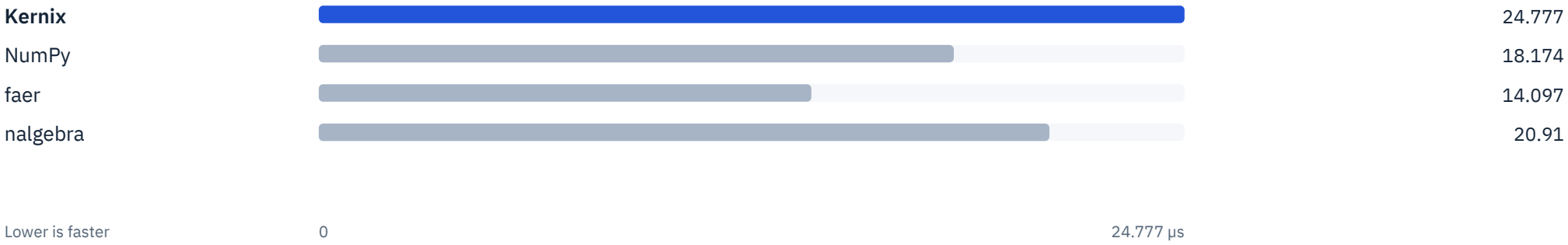
Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



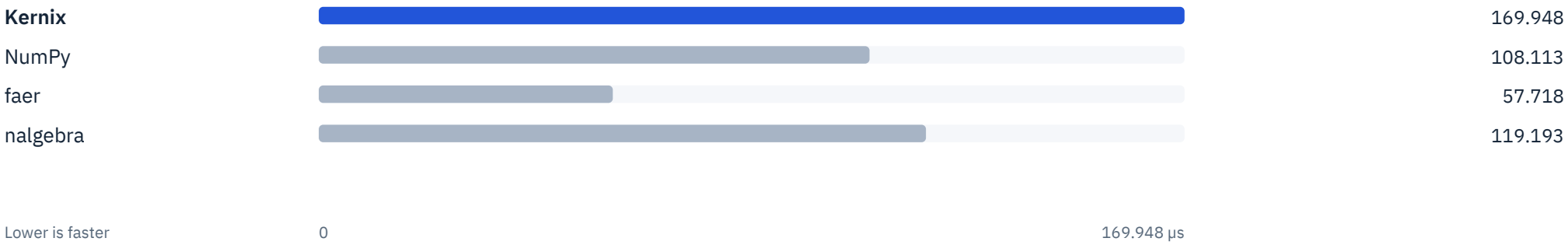
Kernix 0.11.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

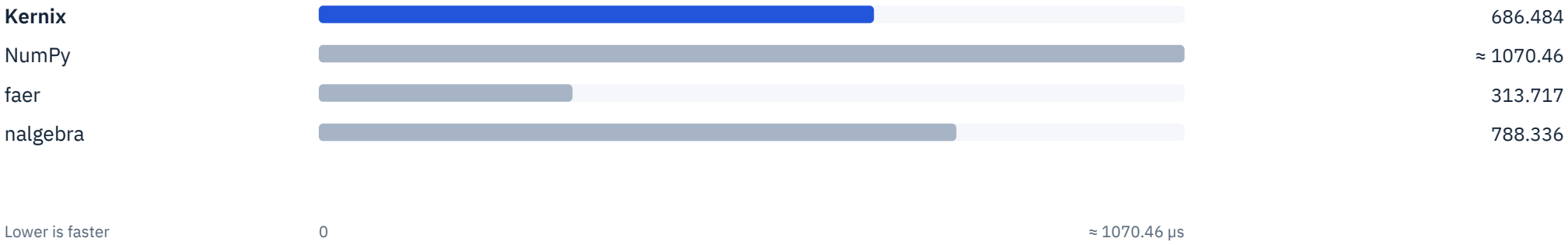
128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



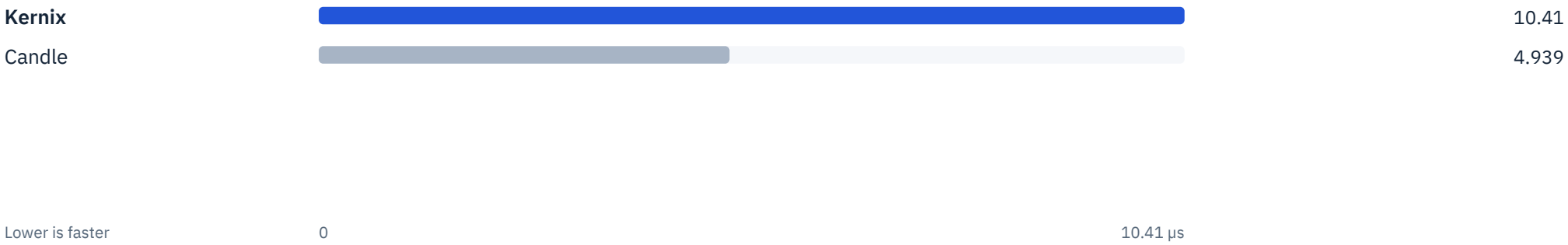
Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median

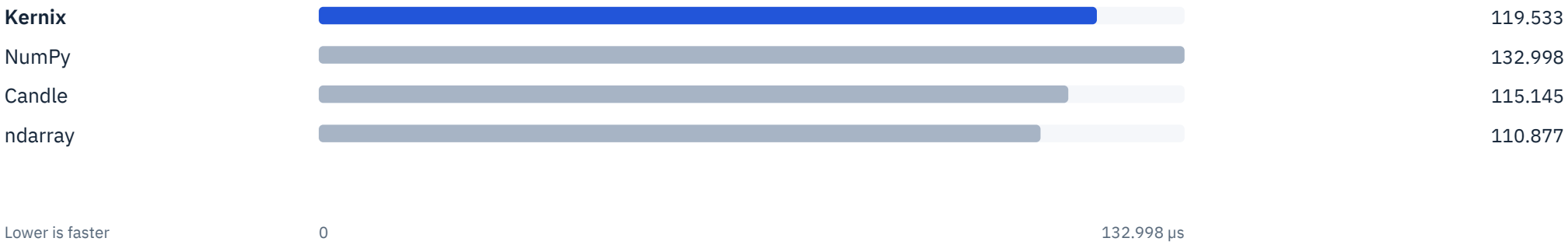


Kernix 0.11.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



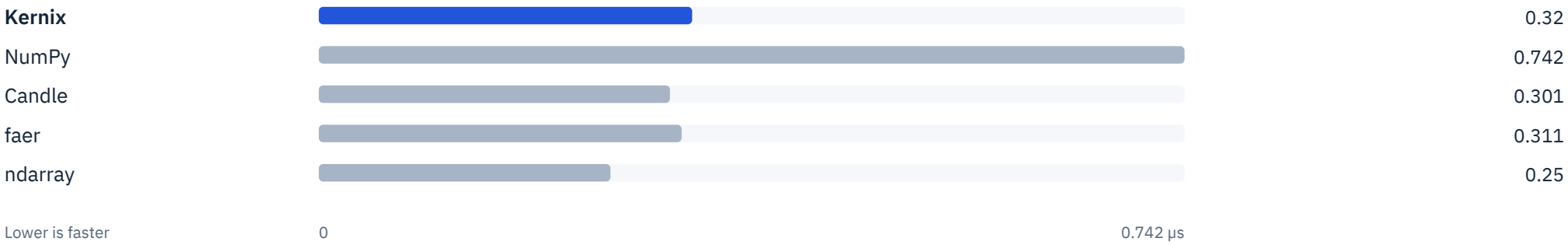
Kernix 0.11.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0009641651936365	0.9952760966204274 – 1.0073507188808772

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0004347259053168	0.9980513997445168 – 1.0019336476416016

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.001448242726027	0.9985006212294634 – 1.0042017208478298

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.999291967226579	0.9974740323731636 – 1.0009405573441617

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9995639040163377	0.9955594464602143 – 1.003105490635505

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.001321604991767	0.9934858354794728 – 1.0058788947677837

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0105549175378568	1.0039669636470052 – 1.0137466886231832

Cholesky factorization

256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.6790232533939629	0.6733664516997802 – 0.6847512849206271

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.5090629428301201	0.5064715577820311 – 0.5154443163666095

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0058258237116304	1.0012331107861963 – 1.019572302490545

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0005125276025437	0.9991821128070911 – 1.0050150348855331

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0015813276827221	0.9912324492979719 – 1.061308636493387

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	94016	94.016
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	102652	102.652
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	49803	49.803
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	45996	45.996
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2214	2.214
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	3677	3.677
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2224	2.224
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2124	2.124
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2043.9999999999998	2.0439999999999998
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2524	2.524

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15509	15.509
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12844	12.844
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12724	12.724
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	14958	14.958
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	15148	15.148
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13455	13.455
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	102031	102.031
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	92934	92.934
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	204954	204.954
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	100428	100.428

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	204002	204.002
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	93986	93.986
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	730869	730.869
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	766815	766.815
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	904363	904.363
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	910935	910.935
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	738492	738.492
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	969514	969.514
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7354	7.354
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6583	6.583

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6512	6.512
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7454	7.454
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6612	6.612
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8105	8.105
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	45836	45.836
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	74700	74.7
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	46186	46.186
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	49011	49.011
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	73688	73.688
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	51556	51.556

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	352229	352.229
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	361626	361.626
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	600354	600.354
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	353150	353.15
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	418683	418.683
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	600013	600.013
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3247	3.247
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4188	4.188
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3126	3.126
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	18966	18.966

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	56566	56.566
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	393688	393.688
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	34084	34.084
Cholesky factorization / nalgebra 512 × 512 · f64	7356338	7356.338
Cholesky factorization / NumPy 512 × 512 · f64	6194474	6194.474
Cholesky factorization / faer 512 × 512 · f64	2584958	2584.958
Cholesky factorization / Kernix 512 × 512 · f64	3886384	3886.384
Column-pivoted QR / Kernix 96 × 32 · f64	78687	78.687
Column-pivoted QR / faer 96 × 32 · f64	42480	42.48
Column-pivoted QR / nalgebra 96 × 32 · f64	78216	78.216

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	343322	343.322
Column-pivoted QR / nalgebra 256 × 64 · f64	767306	767.306
Column-pivoted QR / Kernix 256 × 64 · f64	851052	851.052
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	102482	102.482
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	103403	103.403
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	62255.99999999999	62.25599999999999
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	178364	178.364
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	188583	188.583
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	84257	84.257
Thin QR / Kernix 96 × 32 · f64	67256	67.256

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	54262	54.262
Thin QR / NumPy 96 × 32 · f64	70692	70.692
Thin QR / nalgebra 96 × 32 · f64	58970	58.97
Thin QR / faer 256 × 64 · f64	418203	418.203
Thin QR / Kernix 256 × 64 · f64	780099	780.099
Thin QR / NumPy 256 × 64 · f64	540582	540.582
Thin QR / nalgebra 256 × 64 · f64	570307	570.307
Cholesky factorization / NumPy 64 × 64 · f64	18174	18.174
Cholesky factorization / nalgebra 64 × 64 · f64	20910	20.91
Cholesky factorization / faer 64 × 64 · f64	14097	14.097

Recorded precision / exact latency table

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WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	24777	24.777
Cholesky factorization / Kernix 128 × 128 · f64	169948	169.948
Cholesky factorization / nalgebra 128 × 128 · f64	119193	119.193
Cholesky factorization / faer 128 × 128 · f64	57718	57.718
Cholesky factorization / NumPy 128 × 128 · f64	108113	108.113
Cholesky factorization / faer 256 × 256 · f64	313717	313.717
Cholesky factorization / Kernix 256 × 256 · f64	686484	686.484
Cholesky factorization / nalgebra 256 × 256 · f64	788336	788.336
Cholesky factorization / NumPy 256 × 256 · f64	1070464	1070.464
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4939	4.939

Recorded precision / exact latency table

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WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10410	10.41
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4067	4.067
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7725	7.725
Model inference / Candle batched normalized/sliced MLP · f64	5030	5.03
Model inference / Kernix batched normalized/sliced MLP · f64	3226	3.226
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4098	4.098
Model recording + forward / Candle batched normalized/sliced MLP · f64	6201	6.201
Model backward / Kernix prepared graph → fresh gradients · f64	5099	5.099
Model backward / Candle prepared graph → fresh gradients · f64	20689	20.689
Parameter update / Kernix transactional SGD, fixed gradients · f64	802	0.802

Recorded precision / exact latency table

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WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63067	63.067
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	11331	11.331
Row softmax / Candle 256 × 64 · f64	115145	115.145
Row softmax / ndarray 256 × 64 · f64	110877	110.877
Row softmax / Kernix 256 × 64 · f64	119533	119.533
Row softmax / NumPy 256 × 64 · f64	132998	132.998
Create zeros / Kernix 64 × 64 · f32	320	0.32
Create zeros / faer 64 × 64 · f32	311	0.311
Create zeros / ndarray 64 × 64 · f32	250	0.25
Create zeros / Candle 64 × 64 · f32	301	0.301

Recorded precision / exact latency table

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WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	742	0.742
Create ones / Candle 64 × 64 · f32	471	0.471
Create ones / NumPy 64 × 64 · f32	2094	2.094
Create ones / Kernix 64 × 64 · f32	431	0.431
Create ones / ndarray 64 × 64 · f32	351	0.351
Create ones / faer 64 × 64 · f32	391	0.391